BETHANY HIGH SCHOOL, Sarjapur, Bangalore
PRACTICE WORKSHEET
CH: CHEMICAL CHANGES AND REACTIONS
Std – VIII
CHEMISTRY

Question 1:

(i) State the specific type of chemical reaction represented by each of the following equations:

(a) $Cu(OH)_2 \rightarrow CuO + H_2O \longrightarrow$ Thermal Decomposition

(b) $2S + 3O_2 \rightarrow 2O_3 \longrightarrow$ Synthesis / Direct Combination

(c) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2 \longrightarrow$ Thermal Decomposition

(d) $Cl_2 + 2KI \rightarrow 2KCl + I_2 \longrightarrow$ Single Displacement

(e) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 \downarrow + 2NaCl \longrightarrow$ Precipitation reaction

(f) $CuO + 2HCl \rightarrow CuCl_2 + H_2O \longrightarrow$ Neutralisation reaction

(g) $NH_4Cl \rightleftharpoons NH_3 + HCl \longrightarrow$ Thermal Dissociation

(ii) Complete the following reactions by writing the necessary catalyst, temperature, or state symbols above the arrow:

(a) $N_2 + 3H_2 \rightarrow 2NH_3$ $\quad N_2 + 3H_2 \xrightarrow[450°C-500°C,\ 200\ atm]{Fe\ (Catalyst)\ /\ Mo\ (Promoter)} 2NH_3$

(b) $2H_2O_2 \rightarrow 2H_2O + O_2$ $\quad 2H_2O_2 \xrightarrow{MnO_2\ (Catalyst)} 2H_2O + O_2$

Question 2

(i) Examine the two experimental setups described by the equations below:

Setup 1: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Setup 2: $Cu + H_2SO_4 \rightarrow$ No reaction

Explain why Setup 1 proceeds successfully while Setup 2 fails.

Setup 1: Succeeds, because Zinc lies above hydrogen in the metal reactivity series. It is reactive enough to displace hydrogen from dilute sulphuric acid.

Setup 2: Fails because Copper lies below hydrogen in the reactivity series. It is less reactive and cannot displace hydrogen from the acid.

(ii) When silver nitrate reacts with sodium chloride, a reaction takes place:

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

(a) Break this reaction down into its constituent ions

Constituent Ions:

$AgNO_3 \rightarrow Ag^+ + NO_3^-$

$NaCl \rightarrow Na^+ + Cl^-$

(b) Identify which specific compound form the precipitate.

Silver chloride [AgCl]

Question 3:

(i) All neutralization reactions are double decomposition reactions, but all double decomposition reactions are not neutralization reactions. Defend this statement by comparing the two equations below:

Equation 1: $NaOH + HCl \rightarrow NaCl + H_2O$

Equation 2: $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

- Both equations involve a mutual exchange of ions/radicals, making them both double decomposition reactions.
- **Equation 1:** is a neutralisation reaction because it specifically involves an acid and a base reacting to yield salt and water only.
- **Equation 2:** is a precipitation reaction where two salt solutions react to produce an insoluble solid $BaSO_4$.

(ii) Write a fully balanced chemical equation for each scenario:

(a) Lead nitrate crystals are strongly heated $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

(b) Phosphorus pentoxide reacts with water $P_2O_5 + 3H_2O \rightarrow 2H_3PO_4$